

AI EPC Platform - Engineering Specification

Engineering Specification

Project : Delta Chemical Plant

Vendor : Siemens

Equipment : Chiller

Issue Date : 2025-09-08

Project Overview

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The project consists of the construction of a Tier III data center supporting mission-critical operations.

The project includes utility power integration, UPS systems, cooling systems, and fire protection.

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Cooling System

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CRAH units shall support automatic failover.

Supply air temperature shall remain between 18°C and 27°C.

Each cooling unit shall include vibration monitoring sensors.

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The cooling system shall provide N+1 redundancy.

CRAH units shall support automatic failover.

Cooling equipment shall support SNMP monitoring.

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The cooling system shall provide N+1 redundancy.

Chilled water pumps shall operate with redundant motors.

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Cooling redundancy shall be verified during commissioning.

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Electrical System

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Electrical panels shall comply with IEC standards.

Emergency shutdown shall disconnect non-critical loads.

UPS capacity shall be 500 KVA.

Power Distribution Units shall provide branch circuit monitoring.

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Dual power feeds shall be provided for critical equipment.

Electrical panels shall comply with IEC standards.

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UPS capacity shall be 500 KVA.

Generator backup shall support 72 hours of continuous operation.

Emergency shutdown shall disconnect non-critical loads.

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Fire Protection

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Emergency evacuation procedures shall be displayed.

VESDA detection shall be installed.

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Fire alarm panels shall integrate with the BMS.

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FM200 suppression shall protect all server rooms.

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Network Infrastructure

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Core switches shall support redundant uplinks.

Dual fiber entry points shall be provided.

All network racks shall use Cat6A cabling.

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Network monitoring shall integrate with the NOC.

Fiber optic backbone shall use single-mode fiber.

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Commissioning reports shall include all test results.

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Quality Assurance

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Inspection reports shall be approved before the next activity.

Vendor documentation shall be reviewed before installation.

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All materials shall comply with project specifications.

All construction activities shall follow QA/QC procedures.

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